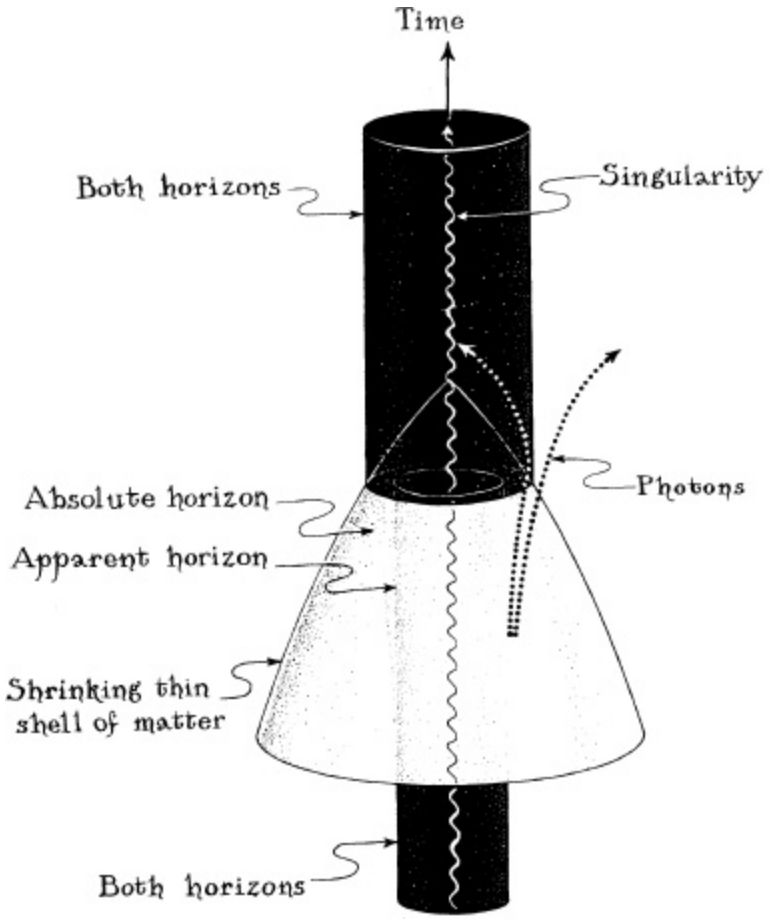


Hawking is a bold thinker. He is far more willing than most physicists to take off in radical new directions, if those directions "smell" right. The absolute horizon smelled right to him, so despite its radical nature, he embraced it, and his embrace paid off. Within a few months, Hawking and James Hartle were able to derive, from Einstein's general relativity laws, a set of elegant equations that describe how the absolute horizon continuously and smoothly expands and changes its shape, in anticipation of swallowing infalling debris or gravitational waves, or in anticipation of being pulled on by the gravity of other bodies.

In November 1970, Stephen Hawking was just beginning to reach full stride as a physicist. He had made several important discoveries already, but he was not yet a dominant figure. As we move on through this chapter, we shall watch him become dominant.